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TorchSpatial: A Location Encoding Framework and Benchmark for Spatial Representation Learning

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SPATIALLY
EXPLICIT AI

Background

Spatial representation learning (SRL):

A set of technique learning neural spatial representations from spatial data in their **native formats** while avoiding manual feature engineering or data conversion.

Application:

Urban planning, environmental monitoring, autonomous navigation, and precision agriculture to analyze spatial patterns, optimize resource allocation, etc.

e.g.,



Sustainability
Wealth Index Prediction
(Sheehan et al., KDD 2019;
Manvi et al., ICLR 2024)

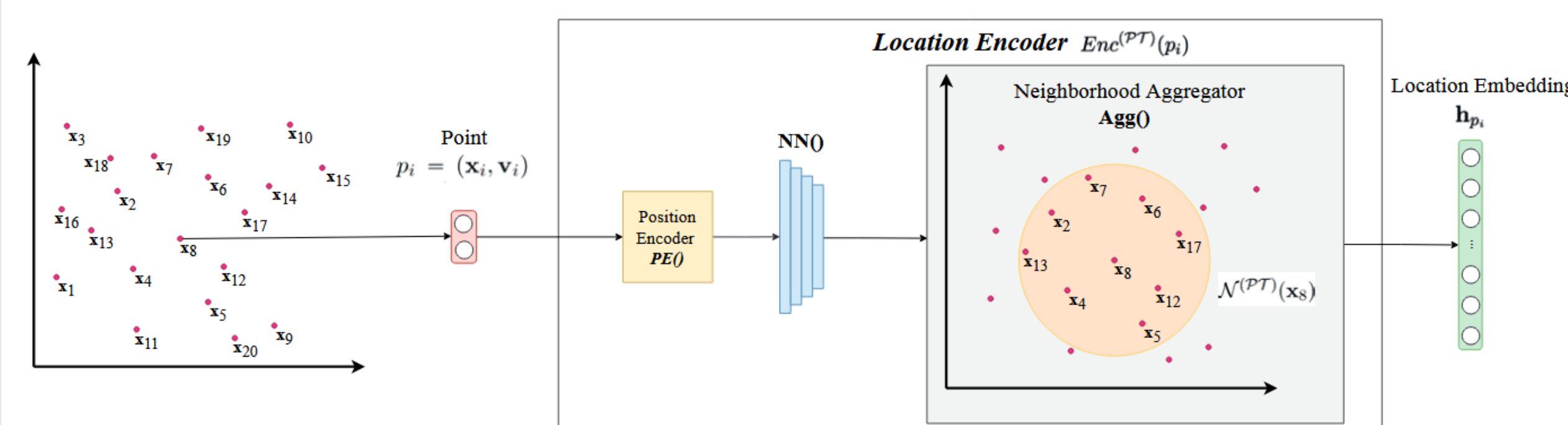
Species Distribution Modeling
iNat2017
(Mac Aoda et al., ICCV 2019; Mai et al., ICLR
2020; Mai et al., Wu et al. NeurIPS 2024)

Motivation

Current challenges:

- **Lack of a community-shared framework** for SRL model development
- **Lack of benchmarks** to systematically evaluate the location encoders' impact on model performance
- **Lack of evaluations** on the geographic bias of geo-aware AI approaches

Location Encoder Framework



Input Location: $\mathbf{x} \in \mathbb{R}^L$ ($L = 2, 3$)

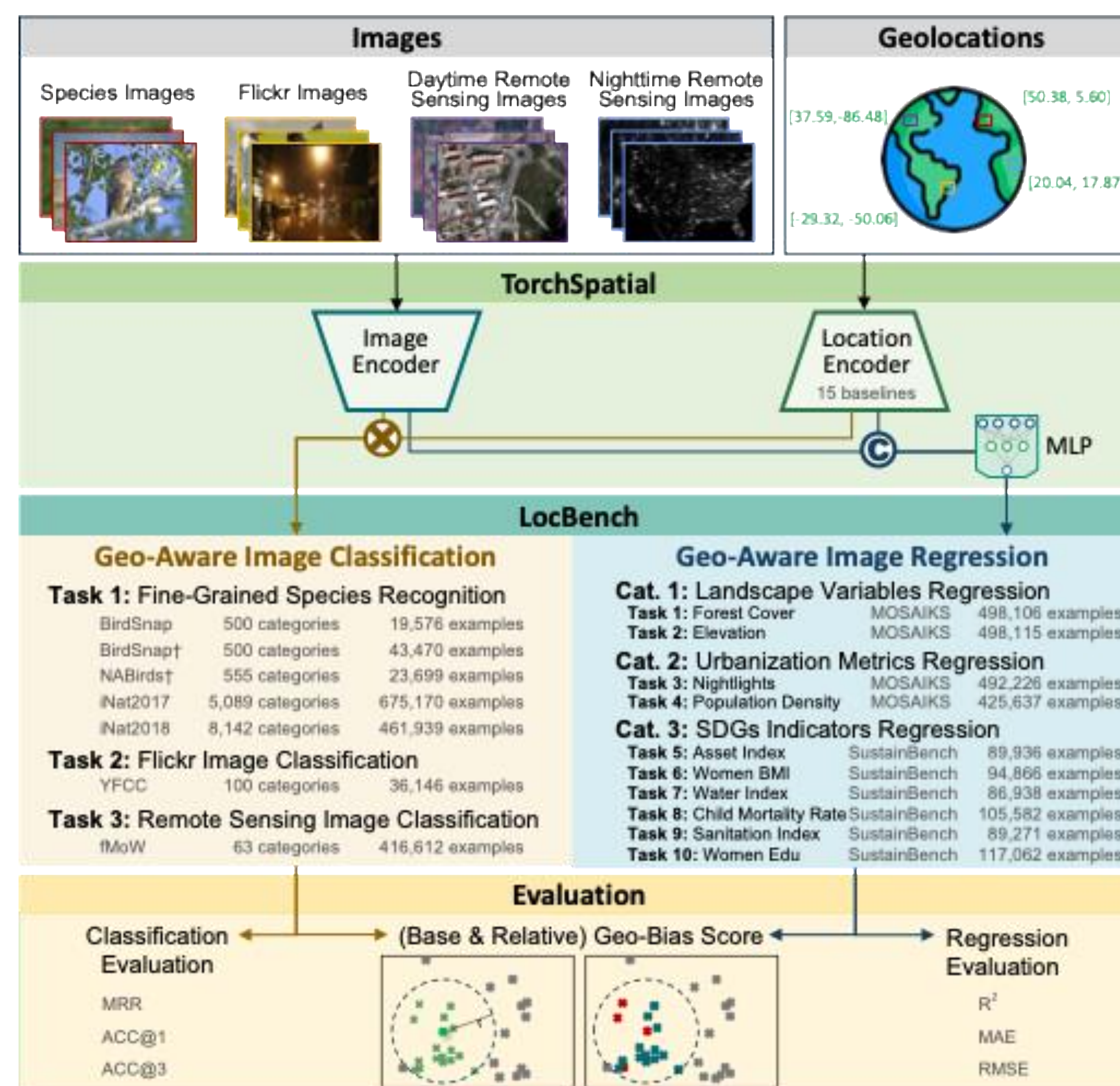
Position Encoder: $PE(\mathbf{x}) \in \mathbb{R}^W$

Learnable Neural Nets: $\mathcal{NN}(\cdot) : \mathbb{R}^W \rightarrow \mathbb{R}^d$

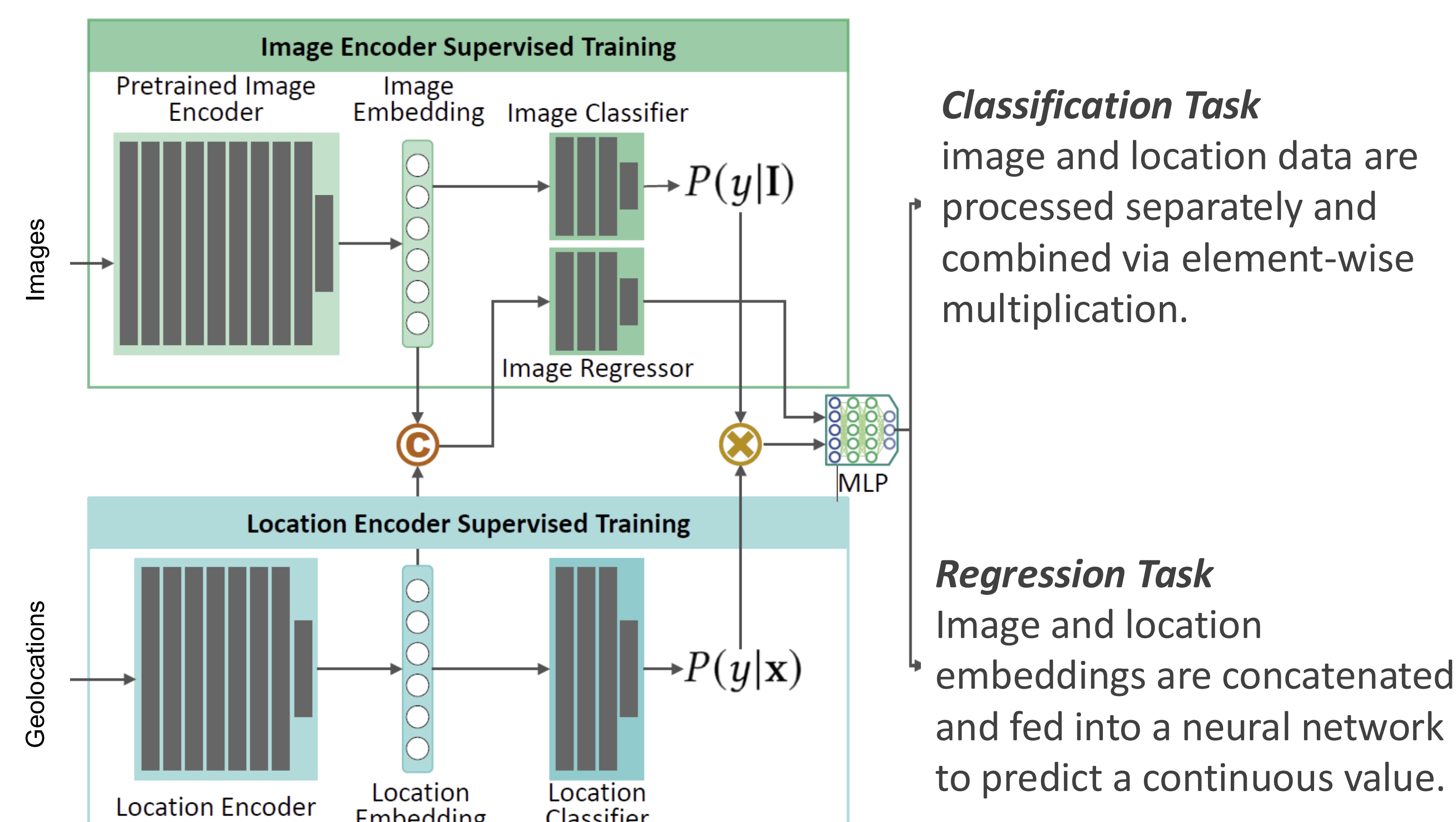
Learnable Neural Nets: $\mathcal{NN}(\cdot) : \mathbb{R}^W \rightarrow \mathbb{R}^d$

A unified, flexible framework that supports the development of location encoders, ensuring scalability and reproducibility.

TorchSpatial



Model Inference Architecture



Classification Task

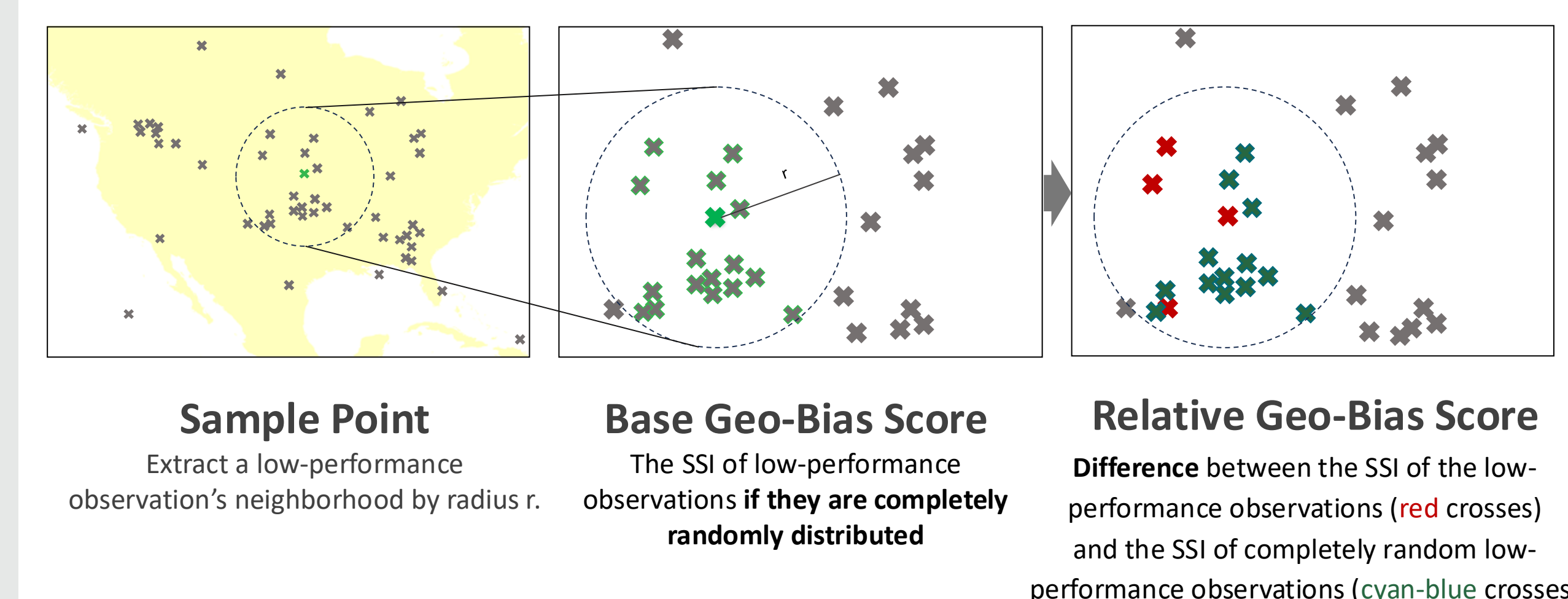
image and location data are processed separately and combined via element-wise multiplication.

Regression Task

Image and location embeddings are concatenated and fed into a neural network to predict a continuous value.

Geo-Bias Score

We employ the **spatial self-information (SSI)**, using a Gaussian distribution to approximate the probability of observing certain types of spatial patterns.



The lower the SSI, the more likely the current spatial patterns arise randomly, namely less geo-biased.

Contribution

- A **location encoder framework** that consolidates 15 location encoders and necessary model building blocks for future location encoders.
- A **benchmark** which contains 7 geo-aware image classification and 10 image regression datasets.
- A **set of evaluation metrics** to quantify location encoders' overall model performance and their geographic bias.
- **All evaluation result** details please refer to the paper.
- The dataset and code are available on our GitHub and website.
- A **Python package** with the same name has been released.



Selected References

- Mac Aodha, Oisín, et al. (2019). "Presence-only geographical priors for fine-grained image classification." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2019.
- Mai, Gengchen, et al. (2022). "A review of location encoding for GeoAI: methods and applications." International Journal of Geographical Information Science 36.4 (2022)
- Wang, Zhangyu et al. (2024). "Probing the Information Theoretical Roots of Spatial Dependence Measures." Proceedings of the 16th Conference on Spatial Information Theory (COSIT)

